

Rodrigo Esparza Giacomelli

AI Software Engineer

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PROFILE

- Software Engineer with 4+ years of experience designing and building scalable systems, currently focused on Artificial Intelligence-based solutions
- Hands-on experience developing Machine Learning and Deep Learning models, from data exploration to production deployment
- Implementation of data and training pipelines, optimizing performance, data quality, and model reproducibility
- End-to-end AI application development: integrating models into APIs, backend services, and real-world products used by users
- Experience working with LLMs and generative models (prompting, fine-tuning, RAG, evaluation), applied to concrete business use cases
- Strong engineering focus: testing, model versioning, monitoring, and MLOps best practices
- Ability to translate business problems into AI technical solutions, collaborating with product, data, and stakeholders
- Strong communication skills with a high level of fluency in English, native proficiency in Spanish, and basic Italian.
- Holding Stamp 1G Visa available to work full time in Ireland. Also Italian citizenship in process.

SKILLS

- **Programming Languages:** Strong experience with **Python** and **SQL** for data manipulation, automation, and backend development (**Pandas**, **NumPy**, **SQLAlchemy**, **Django**). Solid knowledge of OOP, Data Structures, and Algorithm Optimization with **C++** and **Java**. API development and data extraction with **C#**, and experience in **JavaScript/TypeScript** for frontend.
- **Software Development:** Experienced in **Angular** and **React** for frontend, with strong backend skills in **Django**, **MySQL**, and **MongoDB**.
- **Data Analysis & Visualization:** **Python** (data collection, cleaning and manipulation) **SQL** (Subqueries, JOINS, Windows Functions), **Excel** (Power Query, Pivot Tables), business intelligence/visualization tools including **Tableau**, **Matplotlib**, **Seaborn** and **Plotly**
- **AI/Machine Learning:** Experience in **Supervised**, **Unsupervised**, **Reinforcement** and **Learning**, **Neural Networks** (Scikit-learn, TensorFlow, PyTorch), skilled in **NLP** (spaCy, NLTK), **LLM** with RAG architecture and **time series** forecasting and **big data processing** (Dask, PySpark)
- **Cloud & DevOps:** Experience deploying projects on **AWS** and **Azure** for scalable and efficient cloud solutions
- **Version Control:** Skilled in utilizing **Git** and **GitHub** for version management and team collaboration
- **Operating Systems:** Competent in **Windows** and **macOS**, for system maintenance, and server management
- **Management Tools:** Proficient in using **Trello**, **Monday.com**, **Slack**, **Microsoft Teams**, **Notion**, and **Office 365** (Word, Excel, PowerPoint) for project management, collaboration, and documentation

PROFESSIONAL EXPERIENCE

AI Consultant

GRAVITAD

Remote from Dublin, Ireland **Dec 2024 - Dec 2025**

- Designed technology architectures and logical workflows for 10+ startup projects, ensuring scalable and efficient AI integration.
- Selected appropriate AI algorithms and defined necessary implementation variables to optimize feature performance and business outcomes.
- Identified potential risks, improvement opportunities, and business ideas leveraging AI, providing comprehensive technology plans for seamless adoption.

Solution Engineer

MASTTRO

Monterrey, Mexico **Feb 2023 - Jan 2024**

- Promoted to lead architecture decisions for new customer requests and asset implementations, defining scalable and maintainable solutions.
- Implemented automation to download files from 50+ customers by simulating user behavior on web portals, streamlining data collection processes.
- Designed and developed 20+ internal administration web pages, handling end-to-end functionality from **UI/UX** to **backend integration**.
- Architected and implemented an **LLM AI Chatbot** for customer service, capable of handling over 1 million requests per second, enhancing service efficiency and responsiveness.

Data Engineer

MASTTRO

Monterrey, Mexico **Jan 2021 - Feb 2023**

- Responsible for extracting information from multiple sources using **Python**, **C#**, **SQL Server**, and **Azure**, ensuring data integrity and up-to-date information at all times.
- Daily extraction of information from 50+ **APIs** serving over 1,000 users, retrieving up-to-date data and inserting it into the database efficiently.
- Designed and executed **ETL** pipelines to process 1+ million files daily in multiple formats (.txt, .json, .csv), extracting individual content and loading it into the database.
- Automated Office 365 email workflows to handle 50+ customer emails daily with attachments, downloading files and ingesting data into the database seamlessly.
- Developed 100+ **RESTful API** services to support web application frontend, ensuring reliable and scalable data delivery.

Full Stack Developer

DYCSI

Monterrey, Mexico **Jan 2019 - Jan 2020**

- Developed the frontend using Mendix (**HTML/CSS/JavaScript**) and backend with **Django/Python/MySQL** for a banking loan application serving 1,000+ customers.
- Implemented ML algorithms such as **Random Forest** and **Decision Tree** to predict loan repayments and provide personalized loan recommendations.
- Applied **K-Means** clustering to segment clients, automating targeted marketing emails and improving customer engagement.
- Created KPIs and real-time visualizations using **Matplotlib**, **Seaborn**, and **Pandas** to monitor the performance of 5,000+ processes, enabling data-driven decision-making.

Backend Developer

TERNIUM

Monterrey, Mexico Aug 2018 - Dec 2018

- Designed and developed **Django/Python APIs** to provide real-time location tracking for 1,000+ industrial workers using CISCO Network infrastructure, ensuring accurate and reliable data flow.
- Architected and implemented both NoSQL (**MongoDB**) and SQL (**MySQL**) databases, managing over 1 million daily inserts while optimizing performance and data integrity.

EDUCATION

Masters Degree

National College of Ireland (NCI)

Dublin, Ireland Sep 2024 - Sep 2025

- Artificial Intelligence:** Machine Learning | Deep Learning | Neural Networks | NLP | LLMs | Model Training & Prediction | Visualizations | Pandas | Scikit-learn | PyTorch

Masters Degree

Universidad Internacional de la Rioja (UNIR)

La Rioja, Spain Sep 2023 - Mar 2025

- Business & Management:** Strategic Management | Business Transformation | Finance | Operations | Marketing | Sales Strategy | Talent Development | Business Planning

MBA Specialization

City University Miami (MIU)

Miami, United States Sep 2024 - Mar 2025

- Corporate Finance:** Financial Analysis | Risk Management | Capital Structure | Internal & External Financing | Short-Term & Long-Term Instruments | Alternative Financing

Bachelor Degree

Tecnológico de Monterrey (ITESM)

Monterrey, Mexico Aug 2015 - Dec 2020

- Computer Science:** Data Structures | Algorithms | OOP | Software Architecture | Operating Systems | Intelligent Systems | Networking | Cybersecurity | Apps & Web Development | VideoGame Development

PERSONAL TECHNICAL PROJECT

MSc AI Tesis - Real-Time Natural Language Processing and Deep Q-Network for Trading Stock Market

- Aim:** Developed a real-time natural language processing system to analyse international news and forecast their potential impact on the stock market, mixing with Deep Q-Network to predict market behaviour
- Outcome:** Built an end-to-end NLP pipeline capable of ingesting and processing text, applying named entity recognition, sentiment analysis, and custom-trained financial impact models. Integrated with a Deep Q-Network reinforcement learning agent for decision-making based on predicted market reactions.
- Tools and Technologies:** Python (spaCy, Transformers, scikit-learn, pandas, matplotlib), MySQL, Jupyter Notebook and Github

Sales Overlap Analysis

- Aim:** Developed a system to analyze sales overlap among Arca Continental branches, using location data, inventory records, and sales history to identify market cannibalization and improve sales distribution strategies
- Outcome:** Applied machine learning algorithms, including clustering and association rules, to identify recurring customer patterns across branches, resulting in a 10% reduction in financial losses from expired products and improved inventory management
- Tools and Technologies:** Python (scikit-learn, pandas, matplotlib, pyodbc), SQL Server, Jupyter Notebook and Github

Personal Finance Simulator

- Aim:** Developed a personal finance video game to educate students on the impact of daily financial decisions, showing how these choices can affect their long-term financial stability and success
- Outcome:** Developed a web application used in personal finance classes, featuring two user types: players, who can save and restart games with different scenarios, and administrators, who configure scenario outcomes and probabilities.
- Tools and Technologies:** Angular for frontend, Django for backend, MySQL for database management, Github for version control and deployed the application on an Azure server

Economic Impact of COVID-19 in Mexico

- Aim:** Analyzed the economic impact of COVID-19 on Mexico, comparing its performance with other countries to understand the differences and trends in response to the global pandemic.
- Outcome:** Collected, cleaned, and integrated COVID-19 data (national and global), along with performance data from Mexico's top 20 companies and the top 5 companies from diverse countries, into a data warehouse for interactive dashboard analysis
- Tools and Technologies:** Python for data cleaning, MySQL for data management, and Tableau for data visualization

CERTIFICATIONS AND INTEREST

Certifications

- Data Analyst Specialisation, ITESM
- Data Science Specialisation , ITESM
- Angular: From cero to expert, Udemy
- React – The complete Guide 2025, Udemy
- Python Django – Practical Guide, Udemy
- Data Analysis and Visualizations, Udemy

Personal Interest

- Member of a writing club
- Regular reader, one book per month
- Interested in psychology, nutrition, and mindfulness
- My favorite hobby: Chess
- Always seeking to step out of my comfort zone